

COLE CAMPBELL

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SUMMARY

New-grad data scientist and ML engineer with a deep applied portfolio in generative AI, deep learning, and statistical modeling. Builds end-to-end systems, from data pipelines through model integration in production backends. Pursuing an M.S. in Data Science atop a 2026 B.S.

EDUCATION

Arizona State University · Tempe, AZ

M.S., Data Science, Analytics & Engineering, School of Mathematical & Statistical Sciences · *Expected 05/2027*. Focus: Computational Mathematics & Data

B.S., Data Science (Computer Science Concentration) · *Completed 05/2026*. GPA 3.52 / 4.00

Barrett, The Honors College · *08/2023 to 05/2026*

TECHNICAL SKILLS

Languages: Python, R, Dart, C/C++, JavaScript (Node.js), SQL, Bash, MATLAB

ML / AI: Deep learning (CNNs, RNNs/LSTMs, Transformers), generative models (Diffusion, GANs, StyleGAN), LLMs & prompt engineering, LoRA/QLoRA/PEFT, transfer learning, neural style transfer, audio/music generation, model evaluation (FID, CLIP, LPIPS, BLEU, perplexity)

Frameworks: PyTorch (torchvision/torchaudio), TensorFlow, Hugging Face (Transformers, Diffusers, Accelerate), scikit-learn, NumPy, Pandas, SciPy, StatsModels, FastAPI, Flutter/Riverpod

Data Science: Regression & statistical modeling, classification, time series, Bayesian, clustering, longitudinal/panel analysis, feature engineering, A/B testing, causal inference, hypothesis testing, visualization (Matplotlib, Seaborn, Plotly)

Data Engineering: ETL & data pipelines, REST API integration, SQLAlchemy + Alembic, PostgreSQL, Firebase, Docker, CUDA, cloud (AWS, GCP, Azure ML)

Tools: Git, Linux, Jupyter, VS Code, OpenCV, librosa, ParaView/VTK, pytest, Weights & Biases, Kaggle/NBA APIs, REST/OpenAPI

Math: Linear algebra, probability, optimization, numerical methods, stochastic processes, statistical theory

PROJECTS

Multi-Architecture Generative Content Studio · Deep Learning Capstone (2026)

PyTorch · Transformers · Diffusion · GANs · CNNs · RNNs

- Built a unified generative system producing story text, scene illustrations, character portraits, and background music from a single prompt.
- Integrated GPT-2, Stable Diffusion, StyleGAN2/3, VGG16/19, and LSTM into one cohesive pipeline.
- Fine-tuned GPT-2 for narrative generation, entity extraction, and structured scene descriptions; implemented Stable Diffusion with LoRA + ControlNet for composition-controlled scenes.
- Trained StyleGAN2 for character portraits, applying GAN inversion and latent editing for cross-scene consistency; used CNN neural style transfer to unify visual aesthetics.
- Built an LSTM music generator trained on 10k+ MIDI files for mood-aligned audio.
- Evaluated outputs with FID, CLIP score, perplexity, and lexical-diversity metrics (0.97 unique-word ratio on generated narratives); benchmarked DCGAN against Stable Diffusion and measured ~390x higher image sharpness (1306 vs 3.3) for Diffusion, driving the pipeline's generator choice.

Tempo · AI Scheduling Assistant (2026)

FastAPI · Flutter · PostgreSQL · LLM / NLP · Constraint Optimization

- Built a cross-platform scheduling app (Flutter front end, FastAPI back end) that generates alternative day/week plans, compares them, and applies the one the user picks.
- Designed a constraint-based scheduling engine that orders tasks, builds the daily timeline, resolves conflicts, and merges blocks into a coherent schedule.
- Engineered a scenario pipeline supporting generation, diff-based comparison, trade-off analysis, and undo/redo state management.
- Added an "Ask Tempo" conversational layer for natural-language task parsing and schedule interaction.
- Backed by SQLAlchemy + Alembic schema management and Firebase auth; hardened with 360+ backend tests at ~84% coverage.

PSID Panel Dataset Analysis · Longitudinal Economics Project (2025)

Python · Statistical Modeling · Longitudinal Analysis

- Analyzed a 17,000-observation PSID panel (1999 to 2023) to study income and wealth dynamics.
- Built the full pipeline: cleaning, transformation, feature engineering, regression modeling, and visualization.
- Found wealth roughly twice as unequal as income (wealth Gini ~0.85 vs income Gini ~0.45), with the median wealth-to-income ratio rising from 1.25 to 2.05 across the panel.

Scientific Visualization Pipeline (ParaView) · Self-Directed (2026)

ParaView · paraview.simple · VTK · NumPy

- Built a NumPy→VTK dataset (3D Gaussian concentration field + analytic ABC-flow velocity) and a fully Python-scripted ParaView pipeline (paraview.simple).
- Generated volume renderings, Contour isosurfaces, slices, and Stream Tracer streamline tubes, with automated figure and orbit-animation export.

Portfolio & demos: cole-campbell.offerreadynow.us

EXPERIENCE

Handshake AI · Remote

AI Trainer / Technical Evaluator (Contract) · 2026 to Present

- Evaluate and refine AI model outputs across technical domains (coding, STEM, data science), giving expert feedback and corrections to improve model reasoning, correctness, and reliability.

Spinato's Pizzeria · Tempe, AZ

Chef / Runner / Driver · 01/2024 to Present

- Mastered every kitchen position within six months, showing rapid learning and adaptability under high-volume pressure while holding accuracy across detail-critical tasks.